# THE SPECTRAL REPULSION

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Abstract. The collision transform decomposes the centered collision deviation $S^\circ$ over odd Dirichlet characters modulo $b^{\ell+1}$. The squared coefficients $|\hat{S}^\circ(\chi)|^2$ and the squared prime character sums $|P(s,\chi)|^2$ define two probability distributions on the same character space. This paper measures their overlap.

In the tested prime bases, the overlap is well fit by $\rho/n$, where $n$ is the number of odd characters and $\rho \approx 0.6$ is stable across all prime bases tested ($b = 3$ to 37, $n = 3$ to 666). Equivalently, the scaled overlap $n\Omega$ stays near 0.6 across the tested bases. Under the random-relabeling null model, where one spectrum is uniformly permuted relative to the other, the expected overlap is exactly $1/n$. Thus the measured overlap is about 60% of that null-model baseline, and the missing 40% is the spectral repulsion.

The repulsion has consequences. The collision-weighted Goldbach correlation $\sum_{p+q=N} S^\circ(p)\, S^\circ(q)$ has near-zero mean in the tested range. The reflection identity creates a further obstruction: if $S(a) \geq 0$ then $S(m-a) \leq -1$, so both primes in a Goldbach pair cannot lie in the same collision half-group.

## 1. Introduction

The preceding papers recorded two computational patterns: the anti-correlation between $|\hat{S}^\circ(\chi)|$ and $|P(s,\chi)|$ [3], and the double transversality at the level of finite base experiments [4]. Those notes do not prove a general mechanism; they isolate a repeatable numerical pattern.

This paper quantifies the avoidance. The two squared distributions $|\hat{S}^\circ|^2$ and $|P|^2$ on the odd characters have a measurable overlap that decays with the group size. The scaled overlap $n\Omega$ is the statistic studied here.

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## 2. The Overlap

**Definition 1.** For a fixed base $b$ at lag $\ell = 1$, let $n = \phi(b^2)/2$ be the number of odd characters modulo $b^2$. Define the normalized distributions

$$p_c = \frac{|\hat{S}^\circ(\chi_c)|^2}{\sum_j |\hat{S}^\circ(\chi_j)|^2}, \qquad q_c = \frac{|P(s,\chi_c)|^2}{\sum_j |P(s,\chi_j)|^2},$$

where the sums run over odd characters. The *spectral overlap* is

$$\Omega(b,s) = \sum_{c=1}^{n} p_c\, q_c.$$

The quantity $\Omega$ is the raw overlap of the two normalized spectra. The comparison benchmark used in this paper is the *random-relabeling null model*: fix the multisets of weights $\{p_c\}$ and $\{q_c\}$, and choose a uniform random permutation $\pi$ of the channels. Then the relabeled overlap is

$$\Omega_\pi = \sum_{c=1}^{n} p_c\, q_{\pi(c)}.$$

Since $\mathbb{E}[q_{\pi(c)}] = \frac{1}{n}\sum_j q_j = 1/n$, one has

$$\mathbb{E}[\Omega_\pi] = \sum_{c=1}^{n} p_c \cdot \frac{1}{n} = \frac{1}{n}.$$

Thus $1/n$ is the exact expected overlap under random relabeling, regardless of how nonuniform $p$ and $q$ are.

## 3. The Repulsion Ratio

**Definition 2.** The *repulsion ratio* is $\rho(b,s) = n\,\Omega(b,s)$, the scaled overlap at base $b$ and point $s$. If the limit $\rho(s) = \lim_{n\to\infty} \rho(b,s)$ exists, it is the asymptotic repulsion ratio.

In the tested bases, $\rho \approx 0.6$ at $s = 0.5$, with no visible dependence on $b$. If $\rho = 1$, the distributions would be as overlapped as the random-relabeling benchmark. If $\rho = 0$, they would be perfectly orthogonal. The observed value $\rho \approx 0.6$ indicates a 40% deficit relative to the random-relabeling benchmark: in the tested bases, the collision spectrum under-overlaps the prime spectrum.

*Remark* 3. The value $\rho \approx 0.6$ is close to $1/\varphi = (\sqrt{5}-1)/2 \approx 0.618$. Whether this is a coincidence or a structural connection to the golden ratio (which selects the prime 3 in the alignment theory [1]) is an open question.

| $b$ | $n$ | $\Omega$ | $1/n$ | $n\Omega$ | $\Omega/(1/n)$ |
|---|---|---|---|---|---|
| 3 | 3 | 0.1931 | 0.3333 | 0.58 | 0.58 |
| 5 | 10 | 0.0766 | 0.1000 | 0.77 | 0.77 |
| 7 | 21 | 0.0257 | 0.0476 | 0.54 | 0.54 |
| 11 | 55 | 0.0108 | 0.0182 | 0.60 | 0.60 |
| 13 | 78 | 0.0078 | 0.0128 | 0.61 | 0.61 |
| 17 | 136 | 0.0041 | 0.0074 | 0.56 | 0.56 |
| 19 | 171 | 0.0046 | 0.0058 | 0.79 | 0.79 |
| 23 | 253 | 0.0024 | 0.0040 | 0.60 | 0.60 |
| 29 | 406 | 0.0015 | 0.0025 | 0.63 | 0.63 |
| 31 | 465 | 0.0013 | 0.0022 | 0.63 | 0.63 |
| 37 | 666 | 0.0009 | 0.0015 | 0.58 | 0.58 |

TABLE 1. Spectral overlap $\Omega$ at $s = 0.5$ across prime bases. The scaled overlap $n\Omega$ is stable near 0.6. Thus the measured overlap is about 60% of the random-relabeling benchmark $1/n$.

## 4. Effective Support

The Shannon entropy $H = -\sum p_c \log p_c$ measures the spread of a distribution. The effective support is $e^H$: the number of characters that carry most of the distribution's weight.

| $b$ | $n$ | eff($\hat{S}$) | eff($P$) | product | product/$n$ |
|---|---|---|---|---|---|
| 3 | 3 | 2.0 | 2.5 | 5.1 | 1.7 |
| 7 | 21 | 7.0 | 10.7 | 74.6 | 3.6 |
| 13 | 78 | 18.0 | 52.0 | 933 | 12.0 |
| 19 | 171 | 33.2 | 112 | 3,716 | 21.7 |
| 29 | 406 | 64.0 | 295 | 18,877 | 46.5 |
| 37 | 666 | — | — | — | — |

TABLE 2. Effective support of the collision spectrum and the prime spectrum. The collision spectrum concentrates; the prime spectrum spreads. Their product grows faster than $n$.

The collision invariant concentrates its Fourier energy on a small number of characters, with effective support growing much more slowly than the ambient dimension in the displayed data. The prime distribution spreads its energy across most characters in the same data. The

two spectra therefore occupy visibly different regions of character space in the tested bases.

## 5. The Goldbach Null

The spectral repulsion has implications for the collision invariant's interaction with additive prime problems.

**Definition 4.** The *collision-weighted Goldbach correlation* is

$$H(N) = \sum_{\substack{p+q=N \\ p,q \text{ prime}}} S^\circ(p)\, S^\circ(q).$$

**Observation 5** (Goldbach null)**.**
The normalized correlation $H(N)/R(N)$ (where $R(N)$ is the Goldbach count) averages to approximately zero across all tested bases.

| $b$ | mean $H/R$ | std $H/R$ |
|---|---|---|
| 3 | +0.000 | 0.63 |
| 5 | +0.001 | 0.89 |
| 7 | +0.002 | 1.20 |
| 10 | +0.002 | 1.55 |
| 13 | +0.004 | 1.75 |

Table 3. The collision-weighted Goldbach correlation averages to zero. Even $N$ from 1,000 to 100,000.

The null result is consistent with the same separation seen in the overlap statistic: the collision-weighted Goldbach correlation does not show a sustained mean in the tested range.

## 6. The Reflection Obstruction

The reflection identity creates a further constraint on collision-class Goldbach problems.

**Proposition 6.** *If $S(a) \geq 0$, then $S(m-a) \leq -1$. Therefore, for any Goldbach pair $p+q=N$ with $p \equiv a \pmod{m}$ and $q \equiv N-a \pmod{m}$, the two primes cannot both have non-negative collision deviation when $N \equiv 0 \pmod{m}$.*

*Proof.* By the reflection identity [1], $S(a)+S(m-a) = -1$. If $S(a) \geq 0$, then $S(m-a) = -1 - S(a) \leq -1$. When $N \equiv 0 \pmod{m}$, the residue of $q$ is $m-a \pmod{m}$, so the two deviations have opposite signs. □

*Remark* 7. For general $N$, the residue of $q$ is $(N-a) \bmod m \neq m-a$ unless $N \equiv 0$. The obstruction is partial, not total: it affects $1/m$ of all even $N$. But it demonstrates that the bilateral symmetry of the collision invariant is structurally incompatible with same-class Goldbach restrictions.

## 7. Remarks

The spectral repulsion quantifies the central observation of this program: the collision invariant and the prime distribution occupy different parts of character space. The repulsion ratio $\rho \approx 0.6$ is stable across all bases tested and is the main numerical invariant recorded by this note.

The collision invariant does not see the $L$-function zeros in the same way the prime spectrum does (anti-correlation, observed in the earlier note). Its collision-weighted Goldbach correlation has near-zero mean in the tested range. And the double-transversality note suggests the same kind of misalignment across bases. Together these observations point toward a base-specific geometry that is not well captured by standard additive or multiplicative averages alone.

The repulsion ratio $\rho \approx 0.6$ is the quantitative signature of this separation. It measures the degree to which the digit function's geometry is transverse to the prime distribution's analytic structure in the tested family. That this ratio is stable across bases and close to a familiar constant is suggestive, but no theorem identifying its source is proved here.

## References

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